Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	C. Morse	NDS 182, 130 (2022).	14-Sep-2021

Q(β^-)=−3819 *SY*; S(n)=5748 *SY*; S(p)=3317 *SY*; Q(α)=1.017×10^4 *5* 2021Wa16
ΔQ(β^-)=759, ΔS(n)=826, ΔS(p)=854 (2021WA16).
S(2n)=13146 SY *738*, S(2p)=5736 SY *799*, Q(εp)=190 SY *933* (2021WA16).
^{287}Fl has been observed as the product of the ^{244}Pu(^{48}Ca,5n) reaction at JINR (2004OG07) and the ^{242}Pu(^{48}Ca,3n) reaction at JINR (2004OG12) and LBNL (2009ST21); as the α-decay daughter of ^{291}Lv at JINR (2004OG07,2006OG05) and GSI (2016HO09 reassigns one chain tentatively assigned to ^{293}Lv in 2012HO12 to ^{291}Lv); and in a gas-phase chemistry experiment at JINR (2010EI01). Chains of genetically linked nuclei were identified based on the observation of time- and position-correlated α decays, terminated by spontaneous fission. ^{287}Fl was identified as a member of these chains based on the comparison of the observed decay properties of these chains with decay properties of known nuclei.
Other: 1999OG07 identifies two decay chains originating with ^{287}Fl, but these results have not been reproduced.
Half-lives, branching ratios, and α-decay energies in this evaluation have been computed from the individual events listed in the references above. Half-life uncertainties have been computed according to the method of 1984SC13. An additional 10 keV systematic uncertainty is assumed for the α-decay energies, which is added in quadrature to the averaged statistical uncertainty.

^{287}Fl Levels

Cross Reference (XREF) Flags

A ^{291}Lv α decay (19 ms)

E(level)	$T_{1/2}$	XREF	Comments
0	0.40 s *+13−8*	A	%α=100; %SF<5 E(level): Assumed ground state. $T_{1/2}$: From sixteen events.